

Agronomic Insights

The comprehensive analysis reveals several critical insights that fundamentally challenge conventional rice management approaches:

Radiation Dominance During Grain Filling The analysis demonstrates that Radiation₄ (grain filling period) has an extraordinary influence on yield (SHAP: 0.141, $r=0.68$), representing approximately 10 times the impact of early-stage radiation. This finding aligns with rice physiology where 70-80% of final grain weight accumulates during the 35-day post-heading period through current photosynthesis rather than stored carbohydrates. The yield response curve shows a clear threshold effect - yields below 6.0 tons/ha occur when grain filling radiation drops below 16 units, while yields above 6.8 tons/ha require radiation exceeding 20 units.

Growth Stage-Specific Climate Sensitivity Hierarchy The timeline analysis reveals a counterintuitive pattern: early-season temperature management (particularly maximum temperatures during tillering) has disproportionate importance compared to late-season temperature control. This suggests that achieving adequate heat unit accumulation during vegetative growth establishes yield potential, while grain filling becomes primarily radiation-limited rather than temperature-limited. The strong negative correlations between some weather variables (-0.77 to -0.59) indicate that certain climate combinations are mutually exclusive, requiring strategic timing decisions.

Soil Fertility Interaction Networks The soil fertility index (phosphorus × organic matter) shows a clear positive relationship with yield, but the correlation matrix reveals complex interactions. The strong positive correlation between organic matter content and available phosphorus ($r=0.79$) suggests these factors work synergistically through enhanced nutrient cycling and soil buffering capacity. However, the presence of both positive and negative SHAP values for organic matter indicates optimal ranges exist, likely due to nitrogen immobilization effects when C:N ratios become excessive.

Weather Pattern Interdependencies The weather interaction matrix reveals strong correlations between temperature variables across growth stages ($r=0.99-1.00$), indicating that seasonal temperature patterns rather than isolated events drive yield responses. This suggests that farmers must plan for entire seasonal weather trajectories rather than trying to manage individual growth stages in isolation. The negative correlations between early precipitation and late-season temperature

variables suggest competing seasonal patterns that require strategic variety selection.

Predictive Power Hierarchy by Variable Category Meteorological factors explain 91.2% of predictable yield variation when analyzed independently, compared to 88.3% for soil chemistry and only 65.4% for management factors. This hierarchy suggests that while farmers have limited control over weather, understanding and adapting to meteorological patterns offers the greatest potential for yield optimization. The fact that management factors show the lowest independent predictive power indicates their effects are primarily through interaction with weather and soil conditions rather than direct yield impacts.

Specific Recommendations for Farmers

Strategic Planting Date Calculation Calculate optimal transplanting dates by working backward 120 days from your region's historical peak solar radiation period. Use local weather data to identify the 30-day window with highest average daily solar radiation, then count back to determine transplanting timing. Install a simple pyranometer or use local meteorological station data to track daily radiation accumulation. Target minimum 18 MJ/m²/day during grain filling, and be prepared to adjust harvest timing if radiation falls short to allow maximum grain filling completion.

Temperature-Driven Early Season Management Monitor daily maximum temperatures and ensure they consistently exceed 25°C during the first 40 days post-transplanting. Use water depth manipulation as a temperature management tool - maintain 3-5cm depth during cool periods to maximize solar heating, but avoid depths below 2cm which can cause temperature stress. If maximum temperatures fall below 22°C for more than 5 consecutive days, implement protective measures such as windbreaks or temporary covers during critical tillering periods.

Integrated Soil Fertility Management Protocol Target soil organic matter levels between 2.5-3.5% through strategic residue management with C:N ratio control. Apply rice straw or organic residues 45-60 days before transplanting with supplemental nitrogen (25-30 kg N/ha) to achieve C:N ratios below 25:1. Coordinate phosphorus applications (20-30 kg P₂O₅/ha) with organic matter additions to maximize synergistic effects through enhanced nutrient cycling and improved soil buffering capacity.

Dynamic Water Management Strategy Implement growth stage-specific water management based on the climate sensitivity timeline. Use alternate wetting and drying (AWD) during vegetative growth when temperature effects dominate, allowing for better root development and nutrient concentration. Switch to continuous flooding (5-8cm depth) beginning 10 days before heading and maintain throughout grain filling to optimize radiation capture and photosynthetic efficiency during the most yield-critical period.

Variety Selection for Climate Adaptation Choose varieties with 110-130 day maturity specifically to position grain filling during peak radiation months in your location. Prioritize varieties with proven performance under your regional early-season temperature patterns rather than focusing primarily on late-season stress tolerance. Consider the strong temperature correlations across growth stages - varieties that perform well during early cool periods will likely benefit from the same seasonal patterns throughout their growth cycle.

Risk-Based Management Prioritization Focus primary attention on radiation optimization during grain filling (highest risk factor) through field positioning, weed control, and avoiding late-season nitrogen applications that promote excessive vegetative growth. Implement secondary risk management for early-season temperature accumulation through planting date flexibility and microclimate modification. Treat soil organic matter management as a medium-term investment with 2-3 year optimization cycles rather than annual adjustments.

Monitoring and Decision Support Systems Establish simple field monitoring protocols using maximum-minimum thermometers and weekly radiation assessment (sunny/partly cloudy/cloudy scoring). Create decision trees for common scenarios: if cumulative temperature units fall behind during tillering, delay heading through variety selection or nitrogen timing; if grain filling radiation appears insufficient, consider late-season potassium foliar applications (0.5% K_2SO_4) to improve photosynthetic efficiency under suboptimal light conditions.

Economic Risk Assessment Integration Given that radiation optimization can potentially increase yields by 15% and full optimization by 25%, prioritize investments based on local cost-benefit ratios. Radiation timing (variety selection and planting dates) offers the highest return-on-investment as it requires minimal additional inputs. Temperature management and soil fertility optimization require higher initial

investments but provide more stable long-term yield foundations with reduced year-to-year variability.

These recommendations recognize that successful rice production requires understanding and working with natural climate patterns rather than trying to overcome them through intensive inputs alone.